

Device Provisioning Policy

The Grand Marina Hotel — Water Monitoring System

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Created By	Sok Leng Chan
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Approved By	[Leave blank — for client signature]

1. Scope and Purpose

This policy governs the provisioning, management, and decommissioning of HYDROLOGIC devices connected to The Grand Marina Hotel's water monitoring system.

Devices Covered

Device ID	Location	Certificate	Status
HYDROLOGIC-MainBuilding-001	Mechanical Room A	device-001.pem	Active
HYDROLOGIC-PoolSpa-002	Pool Pump House	device-002.pem	Active
HYDROLOGIC-Kitchen-003	Kitchen Mechanical	device-003.pem	Active
[Future devices as deployed]			

2. New Device Onboarding Process

Step 1: Device Registration

Register the device's serial number in two systems: (a) HYDROLOGIC Device Registry — internal tracking system; (b) The Grand Marina Hotel Asset Database — inventory system for compliance. Assign a device ID following the HYDROLOGIC-[LOCATION]-[NUMBER] naming convention. Confirm the device ID does not already exist in either system.

Step 2: Certificate Generation

Generate a digital security credential (TLS client certificate) using the internal Certificate Authority tool: `generate_client_certs.py --device-id HYDROLOGIC-[LOCATION]-[NUMBER]`. The script embeds the device ID as the Common Name. The signing key is stored in a secure Hardware Security Module; only authorized security engineers have access.

Step 3: Certificate Installation

Flash the certificate via the secure USB provisioning station in the IT office. Two-person verification required: Technician A installs the certificate; Technician B independently confirms the digital fingerprint matches the registry record. If they do not match, revoke and restart.

Step 4: Connection Verification

Connect the device to the hotel network. Verify:

- Encrypted data channel (MQTT port 8883) established.
- Data transmits via correct topic:
hydrologic/grandmarina/sensors/[location]/metrics.
- Data appears on the monitoring dashboard within 60 seconds.
- Connectivity test: Send 10 test messages; verify all 10 are received by the broker and displayed.

Step 5: Documentation

Update the HYDROLOGIC Device Registry with: Device ID, physical location, certificate fingerprint, expiration date, provisioning date, and technician names. Notify the Facility Manager that the device is active.

Estimated time for full onboarding: 45 minutes per device.

3. Certificate Rotation Schedule

Current certificate validity period: 1 year

Rotation Schedule:

- Manual rotation triggered 30 days before expiration.
- Automated email alerts at 30, 14, and 7 days before expiration sent to Security Engineering and the Facility Director.

Rotation Procedure:

1. Generate a new credential using `generate_client_certs.py`.
2. Schedule a maintenance window during low-activity hours. Coordinate with the Facility Manager 24 hours in advance.
3. Install new credential via portable provisioning kit. Two-person verification required.
4. Verify reconnection to MQTT broker and data resumption within 60 seconds.
5. Revoke the old credential.
6. Update the HYDROLOGIC Device Registry.

Who is responsible: Security Engineering team (primary); Facility Director (backup).

4. Compromised Device Procedure

If a device is suspected to be compromised:

IMMEDIATE ACTIONS (within 15 minutes):

1. Revoke the device's security credential to block connections.
2. Disconnect the device from the hotel network.
3. Notify the Facility Director. Switch to manual water level/pressure checks until monitoring is restored.

INVESTIGATION (within 24 hours):

1. Review message server logs for unauthorized connections or unexpected data patterns.
2. Assess if water system configuration data was exposed; initiate incident response protocols if necessary.

RECOVERY:

1. Factory reset the device.
2. Generate a new security credential with fresh keys and re-provision the device.

Emergency contact: Grand Marina Facilities Security Operations (24/7).

5. Decommissioning Process

When a device is removed from service:

Step 1: Revoke Certificate

Revoke the certificate immediately and add it to the revocation list.

Step 2: Remove from Monitoring

Remove from the message server's authorized list and the clinical dashboard.
Update alert routing rules.

Step 3: Physical Removal

Factory reset the device to wipe all data. Document return or secure destruction.

Step 4: Documentation

Update the HYDROLOGIC Device Registry (mark as "Decommissioned" with date and reason). Archive 90 days of network logs.

6. Policy Review

This policy will be reviewed:

- Annually
- After any security incident
- When new system expansions are planned

Next review date: July 31, 2027

Signatures

Role	Signature	Date
Prepared by:	Sok Leng Chan	07/31/2026
Reviewed by: Maya Chen, Senior Security Engineer		
Approved by: Marcus Webb, General Manager, The Grand Marina Hotel		